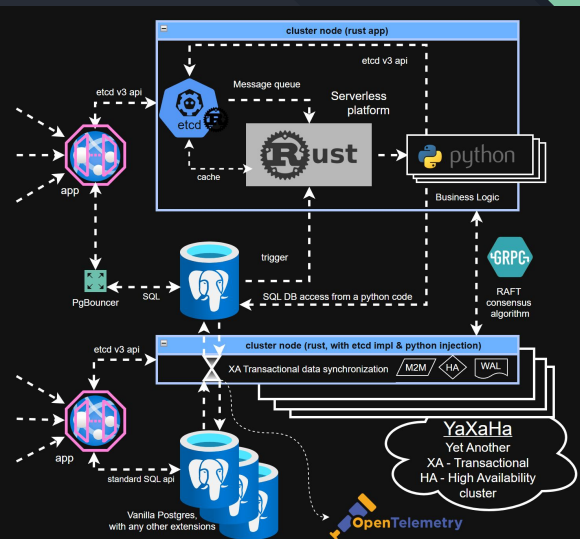


DBinvent

Build your own data source, better than on cloud

Chicago based deep tech startup since 2019



Problems, that many of IT projects faced

01

Data consistency - when transactions are not about money transfer:

AWS: "The second Enactor's clean-up process then deleted this older plan because it was many generations older than the plan it had just applied"
- **this is a transaction isolation issue** which lead to global disaster.

02

Performance on scale-up - when vertical is not an option:

CAP theorem - you can't have either:
Consistency, **A**vailability or **P**artition Tolerance.

03

Backup, restore and repeat - without compromise:

That is possible, but will require a single entry point and additional manual action on failure, and double spending for unused resources.

04

Complex solution exists, but the simple is possible:

No solutions are as simple as RAID or **NAS** for Database which is a heart of many IT projects from just a **website** to distributed commuting.



YaXaHa - The cluster like in cloud, but better and yours

<https://www.dbinvent.com/cluster/> ready to use MVP

01

Strong consistency on write and eventual consistency on read:

Split the performance to a 80% read and 20% write load.

Always able to read, but no write until the data become actual.

02

Virtual dynamic partitioning - the CAP theorem SSC:

Dynamically defining transaction boundaries and spread writing load on different nodes if they are disjoint (patent provisioning) **with AI**.

03

RAFT consensus for a Transaction Coordinator - not a single entry:

The special node called TC act as arbiter to assign workers nodes.

The arbiter does not need to process data.

04

Flexible configuration:

Programmatically defined synchronization rules allows splitting data to redundant for synchronization and ephemeral for a local use.



Roadmap

01

Serverless platform - almost done:

Build your own platform like AWS lambda or Azure Server Function without a vendor lock

02

Web UI - in progress:

Simplify cluster setup by using web UI and provide user-friendly cluster management, including schema changes and testing

03

Testing - in todo as the most priority to prove the durability with a time:

Perform comprehensive consistency, stress and load testing.
Determine SLA and TPS.

04

Sharding and Partitioning - the architecture and analysis phase:

From MPP to Global hierarchical cluster model to substitute client-server interaction and even a mobile API

